

FOUNDATION MODEL RESEARCH & ENGINEERING

8 first-author papers | 5 accepted at NeurIPS · AAAI · ACL · NAACL | 1,003 citations | 2x ACL best awards

EDUCATION

The Hong Kong Polytechnic University, Hong Kong
Ph.D. in Computing · 2023–June 2027
Presidential PhD Fellowship (top 1%)

Fudan University, Shanghai
B.E. in Software Engineering · 2017–2022
GPA rank: top 1% · National Scholarship (top 1%)

EXPERIENCE

Xiaohongshu (REDnote)

Mar 2026 – Present

ACE Talent Program, Foundation Model Team

Scaling Long-Horizon User and Platform Understanding in Foundation Models

Project lead

Problem & objective: Foundation models lack the capabilities to maintain long-horizon user memory and model complex platform dynamics, as well as the diverse domain knowledge required to support XHS production systems across search, recommendation, ads, and other core workflows.

My ownership: Lead the project end to end across task design, data synthesis, rollout strategy, trajectory filtering, evaluation, large-scale CPT and post-training runs, training-data integration, and model release. **(1) TASK SYNTHESIS:** Designed 100+ task templates spanning domains, task types, prediction targets, ground-truth sources, answer formats, and scoring rubrics, scaling data generation to **100K users, 90 domains, 900K tasks, and 200B tokens**. **(2) TASK FILTERING:** Screened synthesized tasks against task-specific constraints, ground truth, and scoring rubrics, removing invalid or ambiguous instances before rollout. **(3) DIFFICULTY-AWARE ROLLOUT:** Ran four student-model rollouts per task and removed tasks that were either too easy or too hard; ran four teacher-model rollouts for each remaining task and retained the highest-scoring trajectory. **(4) TRAJECTORY FILTERING:** Kept only complete, correctly formatted, and verifiable trajectories; filtered bad reasoning patterns such as repeated segments, n-gram repetition, and periodic loops, using an LLM judge for borderline cases. **(5) EVALUATION SYSTEM:** Built frozen in-domain and out-of-domain suites spanning IFBench, AIME, GPQA, MMLU-Pro, and in-house benchmarks; monitored accuracy, response-length distributions, format validity, and repetition to ensure performance gains without introducing degenerate behaviors. **(6) TRAINING & RESULTS:** Observed simultaneous in- and out-of-domain gains, indicating transferable capability gains rather than narrow task fitting. **(7) DATA INTEGRATION:** Integrated final data into the base-model training mixture for release.

Scaling Code-Agent Capabilities through Diverse Environments and High-Quality Trajectories

Core contributor

Problem & objective: Code-agent training cannot scale from task prompts alone; it requires diverse executable environments, automated verification, and a high-throughput rollout stack that reliably produces trainable trajectories.

My ownership: Owned task synthesis and trajectory production, and co-built the company's environment, sandbox, rollout, verification, and evaluation infrastructure. **(1) CODE-AGENT INFRASTRUCTURE:** Co-built sandbox infrastructure and the end-to-end **Sandbox x Harbor x Task x Harness x LLM** pipeline, supporting automated high-concurrency rollouts and model evaluation. **(2) TASK SYNTHESIS:** Built a compositional generator that independently samples nine axes—domain, skill type, primitive skills, persona, language, task complexity, command complexity, fixture type, and verifier type—to create **20K+ executable environments** in Harbor task format. **(3) TRAJECTORY QUALITY:** Rolled out and curated **5K+ verifier-checked trajectories**; removed low-quality or degenerate runs, including incomplete executions, failed verification, and repetitive thinking patterns. **(4) TRAINING RESULT:** Raised Qwen3.5-35B-A3B on Terminal-Bench 2.1 from **44 to 62**, and raised Qwen3.5-397B-A17B from **52.5 to 70** on the same benchmark.

Improving Long-Horizon RSI through Harness Architecture Innovation

First author · Preprint

Problem & objective: Long-horizon agents drift or become trapped in failing strategies, while post-hoc self-improvement cannot recover the current run or immediately validate what it learns.

My ownership: As first author, led the problem formulation, harness architecture, implementation, experimental protocol, trajectory analysis, and paper writing. **(1) HARNESS ARCHITECTURE:** Built a separate supervisor-worker architecture with a two-way live channel: workers notify, ask, and return results; supervisors steer or abort active runs, then distill procedures and failure modes into persistent skills and memory. **(2) ONE-SHOT SETTING:** Across two frozen backbones and three long-horizon benchmarks, PILOT ranks first in 5 of 6 configurations and leads Terminal-Bench 2.0 by up to **9.8 points**. **(3) SELF-IMPROVEMENT SETTING:** Improves pass rate by **12.4–14.6 points**, reduces output tokens by 42.9–47.4%, and increases successful runs per 1M output tokens by **110.3–134.0%**.

EXPERIENCE

Shanghai Innovation Institute

Algorithm Intern

Sep 2025 - Feb 2026

Scaling Agent Capabilities with High-Quality Long-Horizon Data

First author · Preprint

Problem & objective: Agent post-training depends not only on data scale but also on the learning signal carried by each trajectory. Redundant or low-quality demonstrations waste training compute and limit transfer; the objective is to construct high-information, long-horizon interactions that make scaling more effective across model sizes and domains.

My ownership: As first author, led data curation, trajectory generation, cross-scale post-training, evaluation design, ablation analysis, and paper writing.

(1) REAL-WORLD TASKS: Curated authentic vibe-coding and scientific-research requests from professional developers and researchers, preserving long-horizon planning, tool use, and collaboration.

(2) PR-BASED SYNTHESIS: Built a GitHub PR query-synthesis pipeline over 100 repositories with 10K+ stars each; diversified development domains and applied code-change complexity filters and expert review.

(3) TRAJECTORY CURATION: Collected **3.3M high-quality long-horizon trajectory tokens** in the SII CLI environment, preserving complete reasoning, tool calls, environment feedback, error recovery, and human-agent collaboration (42.4K-token average; 152K maximum).

(4) CROSS-SCALE POST-TRAINING: Fine-tuned GLM-4.5-Air (106B) and GLM-4.5 (355B) with a controlled slime-based SFT setup.

(5) EVALUATION SYSTEM: Evaluated AgencyBench, tau2-bench, EvalPlus, DS-1000, and SciCode under both CLI-enabled and tool-free settings to separate tool-use gains from intrinsic capability gains.

(6) RESULTS & INSIGHT: Reached **73.5** on AgencyBench and 57.2 across generalization benchmarks; improved the 106B and 355B base models by 17.3 and 28.4 points, and outperformed a 25M-token SFT baseline by **53.7%**, showing that carefully curated long-horizon data transfers across model scales and domains. [Code](#)

FIRST AUTHOR

PILOT in the Loop: Live Self-Improvement for Long-Horizon Agents

Yang Xiao et al.

Preprint · 2026

A supervisor-worker harness couples live steering with live self-evolution to improve current and future long-horizon runs.

LIMI: Less is More for Agency

Yang Xiao et al.

Preprint · 2025

Agent post-training from high-quality long-horizon trajectories. [Code](#)

LIMOPro: Reasoning Refinement for Efficient and Effective Test-time Scaling

Yang Xiao, Jiashuo Wang, Ruifeng Yuan, Chunpu Xu, Kaishuai Xu, Wenjie Li, Pengfei Liu

NeurIPS · 2025

Improves reasoning accuracy by up to 6.6% while using up to 41% fewer tokens. [Code](#)

SCALE: Selective Resource Allocation for Overcoming Performance Bottlenecks in Mathematical Test-time Scaling

Yang Xiao, Chunpu Xu, Ruifeng Yuan, Jiashuo Wang, Wenjie Li, Pengfei Liu

AAAI · 2026

Allocates inference compute according to sub-problem difficulty, improving Qwen2.5-72B on AIME24 by 35.4 points. [Code](#)

Towards Dynamic Theory of Mind: Evaluating LLM Adaptation to Temporal Evolution of Human States

Yang Xiao et al.

ACL · 2025

Long paper at the ACL main conference.

DataLab: A Platform for Data Analysis and Intervention

Yang Xiao et al.

ACL · 2022

Outstanding Demo Paper Award.

Are All the Datasets in Benchmark Necessary? A Pilot Study of Dataset Evaluation for Text Classification

Yang Xiao et al.

NAACL · 2022

Studies benchmark composition and individual dataset contribution.

How Far Are LLMs from Believable AI? A Benchmark for Evaluating the Believability of Human Behavior Simulation

Yang Xiao et al.

Preprint

Benchmarks the believability of simulated human behavior.

SELECTED CO-AUTHOR

LIMO: Less is More for Reasoning

COLM · 2025

Yixin Ye, Zhen Huang, Yang Xiao et al. · Third author

527 citations and 1.1k GitHub stars. [Code](#)

ExplainaBoard: An Explainable Leaderboard for NLP

ACL · 2021

Co-author

Best Demo Award.

HONORS

PolyU Presidential PhD Fellowship · top 1% · 2023–2027

Outstanding Demo Paper Award, ACL · DataLab · 2022

Best Demo Award, ACL · ExplainaBoard · 2021